SAGAR CHAUDHARY

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SUMMARY

Computer Vision / Perception Engineer with production deep-learning experience across autonomous-driving and ADAS systems. Built multi-modal 3D object detection fusing LiDAR point clouds (SFA3D) with CNN-based camera detections (YOLOv8) using confidence-weighted IoU association - 92.44% / 80.23% AP (easy / hard) on the KITTI benchmark - including LiDAR-to-camera extrinsic calibration via RANSAC/PROSAC and Gaussian/Weighted NMS post-processing. Shipped a production driver foot-monitoring ADAS system combining segmentation, keypoint and object detection with ZoeDepth monocular depth inference. Strong PyTorch, OpenCV and Open3D foundation, with the MLOps skills (Docker, Kubernetes, ONNX Runtime, CI/CD) to take models from benchmark to production.

TECHNICAL SKILLS

Deep Learning: PyTorch, TensorFlow/Keras, CNNs (YOLO, Detectron2, ZoeDepth), LSTM/RNN time-series models, BERT/Transformers, transfer learning, LiDAR-camera sensor fusion, OpenCV, Open3D

Machine Learning: classification, regression, time-series forecasting (LSTM, ARIMA), ensembles (XGBoost, LightGBM, AdaBoost), feature engineering, data cleaning & pre-processing, SMOTE, Optuna, Scikit-Learn, Pandas, NumPy, Statsmodels

MLOps & Deployment: Docker, Kubernetes, KEDA autoscaling, CI/CD (GitHub Actions), Prometheus, OpenTelemetry, ONNX Runtime model serving, structured logging, monitoring, crash-recovery design

Backend & Data Pipelines: Python, Go, Java, FastAPI, REST APIs, WebSockets, TCP/IP sockets, microservices, event-driven pub-sub, Redis Streams, asyncio, JWT, RBAC

Generative AI & Agentic: LangChain, LangGraph, RAG, agentic AI (tool-use agents, MCP), chunking strategies, prompt engineering, prompt-to-template design, rerankers, vector search (ChromaDB), LLM & embedding APIs (OpenAI, Gemini, Ollama, Nomic)

Databases & Cloud: PostgreSQL (SQL), NoSQL (MongoDB), MySQL, ChromaDB, AWS, Linux, Git

Core CS: Data Structures, OOP, Distributed Systems, System Design, Concurrency, Operating Systems, Computer Networks

EXPERIENCE

Deevia Software, Bengaluru — **Apr 2025 - Present**

Graduate Trainee Engineer — *Nov 2025 - Present*

Multi-Vendor Camera Device Management Framework

Tech Stack: Python, FastAPI, PostgreSQL, MongoDB (NoSQL), React, Docker, CI/CD, OOP Design Patterns

- Built a device management framework using object-oriented design - abstract base classes and interface layers - to abstract camera hardware, enabling drop-in integration of multiple camera vendors without backend changes.
- Developed a FastAPI backend exposing REST endpoints for device registration, configuration management, and real-time status monitoring across distributed devices.

Real-Time Engine Test-Bed Monitoring & Telemetry Platform (Industrial IT-OT System)

Tech Stack: Python, FastAPI, LangChain, LangGraph, LSTM (PyTorch), RAG, ChromaDB, Ollama, OpenAI, Gemini, WebSockets, TCP/IP Sockets, PostgreSQL, React.js, TypeScript, Redux Toolkit, Docker, JWT, RBAC

- Architected an end-to-end IT-OT data pipeline for an engine test facility - ingesting high-frequency industrial sensor data over raw TCP/IP sockets and streaming it to live dashboards via WebSockets at sub-second latency.
- Trained LSTM deep-learning models on engine sensor time-series for trend forecasting, anomaly detection, and predictive-maintenance insights during live test runs.
- Designed an event-driven publish-subscribe architecture that decoupled ingestion, processing, and streaming services, so new sensor types are added through configuration rather than code changes.

Machine Learning Intern — *Apr 2025 - Nov 2025*

3D Object Detection via LiDAR-Camera Sensor Fusion (KITTI)

Tech Stack: Python, PyTorch, YOLOv8 (CNN), SFA3D, OpenCV, Open3D, NumPy, SLAM, RANSAC/PROSAC, Docker

- Built a multi-modal 3D object detection system fusing LiDAR point-cloud detections (SFA3D) with CNN-based 2D camera detections (YOLOv8) for robust perception in autonomous-driving scenes.
- Designed a confidence-weighted fusion algorithm that combines per-sensor detections by IoU association, improving accuracy and robustness over single-sensor baselines - 92.44% / 80.23% AP (easy / hard) on the KITTI benchmark.
- Implemented LiDAR-to-camera extrinsic calibration using RANSAC/PROSAC for robust, drift-resistant sensor alignment.
- Built a post-processing pipeline with Gaussian NMS and Weighted NMS to merge overlapping boxes and refine final detections.
- Developed visualization tooling with multi-model attribution and runtime performance statistics for debugging and fusion analysis.

Foot Monitoring System (ADAS)

Tech Stack: Python, PyTorch, YOLOv11, Detectron2, ZoeDepth, OpenCV, FastAPI, React, PostgreSQL, AWS

- Built a production computer-vision system detecting driver pedal interactions (accelerator, brake, clutch) by combining segmentation, keypoint detection, and object detection deep-learning models.
- Developed pedal-position estimation and monocular depth inference using ZoeDepth with custom logic to classify pedal depression and distinct foot actions.
- Built a FastAPI backend with REST endpoints for data access, annotation workflows, and real-time inference, with abstract interfaces for multi-camera modularity.
- Automated data collection, labeling, and continuous model retraining using Dockerized microservices and open-source annotation tools (LabelMe, LabelImg).

Distributed Multi-Sensor Video Analytics Platform

Tech Stack: Go, Python, FastAPI, REST APIs, WebSockets, AWS, PostgreSQL, Docker, Microservices, OpenCV

- Implemented timestamp synchronization to align video frames with multi-modal sensor streams (accelerometer, velocity, gyroscope) for precise temporal correlation.
- Built a multi-format ingestion pipeline supporting heterogeneous camera systems with mixed inputs (video, CSV, binary sensor files).

PROJECTS

Inferno - Distributed ML Inference & Agentic AI Platform (Live Demo · GitHub)

Tech Stack: Python, FastAPI, Redis (Streams + Pub/Sub), WebSockets, asyncio, PyTorch, ONNX Runtime, YOLOv8, faster-whisper, Sentence-Transformers (RAG), MCP (Agentic AI), Prometheus, OpenTelemetry, React, TypeScript, Docker, Kubernetes, KEDA, GitHub Actions

- Built a horizontally-scalable model-serving platform - a stateless async FastAPI gateway, a Redis-Streams broker, and a pool of shared-nothing worker processes - that batches inference requests and streams results back over WebSockets to a live ops dashboard, solving the core serving problem of high throughput and low-latency real-time delivery.
- Applied MLOps end to end - centralized config/contracts spine (pydantic-settings), custom Prometheus collector, OpenTelemetry tracing, structured logging, a Dockerized stack, Kubernetes manifests with KEDA queue-depth autoscaling, green GitHub Actions CI, and a live production deployment.
- Generalized the system with a config-driven, model-agnostic plugin registry serving 7 model types - YOLOv8 detection, Whisper transcription, ONNX/HuggingFace classification, and RAG question-answering.

Disaster Tweet Classification

Tech Stack: Python, NLTK, Scikit-Learn, TensorFlow/Keras, BERT, TF-IDF

- Collected real-time tweet data via the Twitter API and preprocessed noisy social text with tokenization, lemmatization, and stopword removal to build a clean training corpus.
- Benchmarked standalone TF-IDF and BERT models, then built a hybrid classifier combining BERT embeddings with TF-IDF features - 92.8% accuracy / 91% F1, outperforming single-model baselines.

ACHIEVEMENTS & CERTIFICATIONS

- Top 10 Finalist - HumanAIze YouData.ai Hackathon (out of 150 teams); built a scalable NLP pipeline for document classification. Hack2skill certificate (ID: 2024H2S06EH10008).
- 5th Place - CODE RED 2024 Hackathon (24-hour event); led a team of 4 to build a marine-life classification system using CNNs, real-time image processing, and transfer learning.
- Machine Learning Specialization - Stanford / DeepLearning.AI (Coursera), Dec 2023
- Neural Networks and Deep Learning - DeepLearning.AI, May 2024 · OpenCV Bootcamp - OpenCV University, Apr 2025 (Grade: 100%)

EDUCATION

Bangalore Institute of Technology — B.E. in Information Science and Engineering (CGPA: 8.6) **Dec 2021 - Jul 2025**

DAV Public School, Jamshedpur — CBSE Class 12: 92.4% **Apr 2020 - May 2021**

DAV Public School, Jamshedpur — CBSE Class 10: 92.4% **Apr 2018 - Apr 2019**